

# AP-E15 Relaxation Defect and Fibre-Phase Purification: What Is Now Proved and Where the Hopf-Base Blocker Lives

Route-F derivation ledger

23 July 2026

## Abstract

This note freezes the AP-E11 action and replaces the preceding sequence of lattice refinements by one continuum defect problem. If a smooth fixed-degree minimizing sequence converges weakly in the complete-minor graph space, the quadratic derivative-energy measures have the canonical decomposition

$$\frac{1}{2}|\Xi_j|^2 dx \xrightarrow{*} \frac{1}{2}|\Xi_u|^2 dx + \mu, \quad \mu \geq 0.$$

The total mass of  $\mu$  is exactly the local relaxation gap, and  $\mu = 0$  is equivalent to strong complete-minor graph convergence.

For the Hopf split  $n = h(u)$ ,  $a = 2u^*\theta$ , compact fibre changes  $u \mapsto e^{i\chi}u$  send  $a$  to  $a + 2d\chi$  and preserve both the base and degree. On every ball of radius  $r$  satisfying  $1 - m^2r^2/\pi^2 > 0$ , the exact phase functional is strongly convex. Hence an asymptotically minimizing recovery sequence has zero local phase-relaxation drop. In a precise sense, every compact-fibre-phase-reducible vertical defect is therefore zero. This eliminates the AP-E14 rank-one fibre oscillator, but not mixed defects carried by simultaneous oscillation of the Hopf base.

The weighted cutoff quantity left by AP-E13,

$$r^{-2} \int_{B_r(x)} |u - u(x)|^2 |M_2(Du)|^2,$$

is proved to vanish at almost every Lebesgue point of every fixed graph map. That result is sharp in logical scope: it does not interchange the shrinking radius with the recovery-sequence limit. A singular defect can be supported on the exceptional set, while a diffuse base/mixed defect is not yet linked to the fixed-map estimate. The remaining theorem must establish sequence-uniform decay at  $\mu$ -almost every point, or an excess-decay estimate which controls both limits. Full  $\mu = 0$ , local recovery, regularity, isolation, the bosonic Hessian, determinant promotion, and the Route-E portal therefore remain fail-closed. No new lattice scan is introduced.

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# 1 Frozen action and admissible relaxation

Let  $\Omega \subset \mathbb{R}^3$  be bounded, let  $e_0 = (1, 0, 0, 0)$  be the boundary vacuum, and fix degree  $B = 1$ . From AP-E11 onward the only admissible continuum density is

$$W(F) = \frac{1}{2}|F|^2 + \frac{R}{2}|M_2(F)|^2 + \frac{K}{2}|M_3(F)|^2, \quad V(u) = m^2(1 - u^0), \quad (1)$$

with production normalization

$$(R, K, m^2) = (1, 0.35, 0.12). \quad (2)$$

No coefficient, stencil, triangulation, anchoring rule, or new lattice sequence is varied in AP-E15.

Define the complete-minor graph class

$$\mathcal{G}_B = \left\{ u : \Omega \rightarrow \mathbb{S}^3 : \begin{array}{l} u - e_0 \in W_0^{1,2}, \quad M_2(Du), M_3(Du) \in L^2, \\ (2\pi^2)^{-1} \int_{\Omega} \det[u, Du] \, dx = B \end{array} \right\}. \quad (3)$$

For smooth admissible maps put

$$\mathcal{E}(u) = \int_{\Omega} (W(Du) + V(u)) \, dx. \quad (4)$$

Its fixed-degree lower-semicontinuous relaxation is

$$\bar{\mathcal{E}}(u) = \inf_{\substack{u_j \in C^\infty(\bar{\Omega}, \mathbb{S}^3) \\ u_j|_{\partial\Omega} = e_0, \deg u_j = B \\ u_j \rightarrow u \text{ in } L^2}} \liminf_{j \rightarrow \infty} \mathcal{E}(u_j), \quad (5)$$

where sequences of unbounded energy are irrelevant. AP-E11 identifies this relaxation as the regulator limit. It does not identify  $\bar{\mathcal{E}}$  with the naive integral (4).

# 2 The canonical relaxation defect measure

Introduce the Hilbert-valued complete-minor vector

$$\Xi(F) = (F, \sqrt{R} M_2(F), \sqrt{K} M_3(F)), \quad W(F) = \frac{1}{2} |\Xi(F)|^2. \quad (6)$$

Let  $u_j$  be a recovery sequence for a fixed  $u \in \mathcal{G}_B$ , and pass to a subsequence such that

$$u_j \rightarrow u \quad \text{strongly in } L^2, \quad \Xi(Du_j) \rightharpoonup \Xi(Du) \quad \text{weakly in } L^2. \quad (7)$$

The second convergence is an explicit hypothesis: it says that the weak limits of the minors are the minors of the selected graph representative.

**Theorem 2.1** (Defect-measure decomposition). *Under (7), after a further subsequence there is a unique nonnegative Radon measure  $\mu$  on  $\bar{\Omega}$  such that*

$$\frac{1}{2} |\Xi(Du_j)|^2 \, dx \xrightarrow{*} \frac{1}{2} |\Xi(Du)|^2 \, dx + \mu. \quad (8)$$

If  $u_j$  is an exact recovery sequence, then

$$\mu(\bar{\Omega}) = \bar{\mathcal{E}}(u) - \mathcal{E}_{\text{naive}}(u), \quad \mathcal{E}_{\text{naive}}(u) = \int_{\Omega} (W(Du) + V(u)) \, dx. \quad (9)$$

Moreover,

$$\mu = 0 \iff \Xi(Du_j) \rightarrow \Xi(Du) \text{ strongly in } L^2 \quad (10)$$

for that recovery subsequence.

*Proof.* The bounded positive measures  $\frac{1}{2}|\Xi(Du_j)|^2 dx$  have a weak-star convergent subsequence. For every nonnegative  $\varphi \in C(\bar{\Omega})$ , weak lower semicontinuity applied to  $\sqrt{\varphi}\Xi(Du_j)$  gives

$$\frac{1}{2} \int_{\Omega} \varphi |\Xi(Du)|^2 dx \leq \liminf_j \frac{1}{2} \int_{\Omega} \varphi |\Xi(Du_j)|^2 dx.$$

The difference between the limiting measure and the absolutely continuous measure on the left is therefore a nonnegative Radon measure  $\mu$ . Because  $V$  is linear and bounded on  $\mathbb{S}^3$ , strong  $L^2$  convergence implies  $\int V(u_j) \rightarrow \int V(u)$ , proving (9).

If  $\mu = 0$ , testing (8) by one gives convergence of the Hilbert norms. Together with weak convergence this implies strong convergence. The converse follows by taking the strong limit in every weighted quadratic integral.  $\square$

**Remark 2.2** (What  $\mu = 0$  does and does not mean). *For the selected map,  $\mu = 0$  closes the local Lavrentiev/recovery gap and gives complete-minor graph recovery. It does not, by itself, prove continuity, the Euler–Lagrange equation, partial regularity, or isolation. Those require a local comparison theorem and an excess-decay argument.*

### 3 Exact Hopf fibre functional

Write  $u = (z, w) \in \mathbb{S}^3 \subset \mathbb{C}^2$ , let  $n = h(u) \in \mathbb{S}^2$ , and let  $a = 2u^*\theta$ . With the AP-E14 orientations,

$$\begin{aligned} |Du|^2 &= \frac{1}{4}(|a|^2 + |Dn|^2), \\ |M_2(Du)|^2 &= \frac{1}{16}(|a \wedge Dn|^2 + |da|^2), \\ |M_3(Du)|^2 &= \frac{1}{64}|a \wedge da|^2. \end{aligned} \tag{11}$$

For a compact fibre phase,

$$T_\chi u = e^{i\chi}u, \quad n_\chi = n, \quad a_\chi = a + 2d\chi, \quad da_\chi = da. \tag{12}$$

If  $\chi$  has zero trace on a ball, the transformation preserves the exterior trace and the degree because

$$(a + 2d\chi) \wedge d(a + 2d\chi) - a \wedge da = 2d(\chi da). \tag{13}$$

For a smooth  $u$  on  $B_r$  define

$$\begin{aligned} \mathcal{F}_u(\chi) &= \int_{B_r} \left\{ \frac{1}{8}(|a + 2d\chi|^2 + |Dn|^2) \right. \\ &\quad + \frac{R}{32}(|(a + 2d\chi) \wedge Dn|^2 + |da|^2) \\ &\quad \left. + \frac{K}{128}|(a + 2d\chi) \wedge da|^2 + m^2(1 - \Re(e^{i\chi}z)) \right\} dx \end{aligned} \tag{14}$$

on  $H_0^1(B_r)$ . Smoothness of  $u$  makes all coefficients bounded, so this functional is well-defined, coercive, and weakly lower semicontinuous.

**Theorem 3.1** (Local fibre-phase purification). *Set*

$$\lambda_r = 1 - \frac{m^2 r^2}{\pi^2}. \tag{15}$$

*If  $\lambda_r > 0$ , then  $\mathcal{F}_u$  is  $\lambda_r$ -strongly convex with respect to the seminorm  $\|d\chi\|_{L^2}$ . It has a unique minimizer  $\chi_u \in H_0^1(B_r)$  and*

$$\mathcal{F}_u(0) - \mathcal{F}_u(\chi_u) \geq \frac{\lambda_r}{2} \|d\chi_u\|_{L^2(B_r)}^2. \tag{16}$$

*For (2), every  $r < \pi/\sqrt{0.12} = 9.0689968\dots$  is admissible.*

*Proof.* For  $\phi \in H_0^1(B_r)$ , exact differentiation gives

$$D^2\mathcal{F}_u(\chi)[\phi, \phi] = \int_{B_r} \left\{ |\mathrm{d}\phi|^2 + \frac{R}{4} |\mathrm{d}\phi \wedge Dn|^2 + \frac{K}{16} |\mathrm{d}\phi \wedge da|^2 + m^2 \Re(e^{i\chi} z) \phi^2 \right\} \mathrm{d}x. \quad (17)$$

Since  $|z| \leq 1$ , the last term is bounded below by  $-m^2\phi^2$ . The exact first Dirichlet eigenvalue of the three-ball is  $\lambda_1(B_r) = \pi^2/r^2$ , hence

$$\int_{B_r} \phi^2 \mathrm{d}x \leq \frac{r^2}{\pi^2} \int_{B_r} |\mathrm{d}\phi|^2 \mathrm{d}x. \quad (18)$$

Dropping the two nonnegative wedge terms proves

$$D^2\mathcal{F}_u(\chi)[\phi, \phi] \geq \lambda_r \|\mathrm{d}\phi\|_2^2.$$

The direct method gives existence; strong convexity gives uniqueness and the standard Bregman inequality (16).  $\square$

## 4 The vertical defect which really vanishes

Let

$$I_B = \inf\{\mathcal{E}(v) : v \in C^\infty(\overline{\Omega}, \mathbb{S}^3), v|_{\partial\Omega} = e_0, \deg v = B\}, \quad (19)$$

and let  $u_j$  be a smooth sequence with  $\mathcal{E}(u_j) \rightarrow I_B$ . On an interior ball  $Q = B_r(x)$  with  $\lambda_r > 0$ , define the compact-phase relaxation drop

$$\Delta_j(Q) = \mathcal{F}_{u_j}(0) - \inf_{\chi \in C_c^\infty(Q)} \mathcal{F}_{u_j}(\chi) \geq 0. \quad (20)$$

**Theorem 4.1** (Zero compact-phase-reducible vertical defect). *For every fixed admissible ball  $Q$ ,*

$$0 \leq \Delta_j(Q) \leq \mathcal{E}(u_j) - I_B \longrightarrow 0. \quad (21)$$

*If  $\chi_j$  is the unique  $H_0^1(Q)$  phase minimizer, then*

$$\|\mathrm{d}\chi_j\|_{L^2(Q)}^2 \leq \frac{2\Delta_j(Q)}{\lambda_r} \longrightarrow 0. \quad (22)$$

*Consequently the outer defect obtained by covering sets with admissible balls and assigning each ball the asymptotic cost  $\limsup_j \Delta_j(Q)$  is identically zero. Equivalently, every component of  $\mu$  which can be exposed by a compact fibre-phase energy reduction is zero.*

*Proof.* For any  $\chi \in C_c^\infty(Q)$ , extend it by zero and set  $v_j = e^{i\chi}u_j$ . Equation (13) shows that  $v_j$  has the same boundary value and degree as  $u_j$ . Therefore

$$I_B \leq \mathcal{E}(v_j) = \mathcal{E}(u_j) - (\mathcal{F}_{u_j}(0) - \mathcal{F}_{u_j}(\chi)).$$

Take the supremum over smooth compact phases to obtain (21). Smooth phases are dense for the functional with the fixed smooth coefficients of  $u_j$ , so the infimum equals the  $H_0^1$  minimum. Equation (22) follows from (16). Since every fixed admissible ball has zero asymptotic cost and such balls form a countable basis, the associated covering outer measure is zero.  $\square$

**Remark 4.2** (No overclaim about the full vertical-looking energy). *The theorem eliminates pure exact fibre oscillation, including the AP-E14 rank-one phase oscillator with fixed Hopf base. It does not prove strong convergence of  $a_j$ ,  $a_j \wedge Dn_j$ , or  $a_j \wedge da_j$  when  $n_j$  and its curvature oscillate simultaneously. Such a sequence can be stationary along each fibre orbit while retaining base/mixed concentration. Calling that remaining mass “vertical defect” would obscure the actual blocker; hence AP-E15 reserves that name for the phase-reducible component proved zero above.*

## 5 Weighted cutoff decay on the Hopf base

The AP-E13 replacement leaves the dimensionless cutoff defect

$$\mathfrak{D}_u(x, r; q) = \frac{1}{r^2} \int_{B_r(x)} |u - q|^2 |M_2(Du)|^2 dy. \quad (23)$$

Using (11),

$$\mathfrak{D}_u(x, r; q) = \frac{1}{16r^2} \int_{B_r(x)} |u - q|^2 (|a \wedge Dn|^2 + |da|^2) dy. \quad (24)$$

**Theorem 5.1** (Almost-everywhere weighted decay). *For every fixed  $u \in \mathcal{G}_B$ , at almost every  $x \in \Omega$ , with the precise representative  $q = u(x)$ ,*

$$\boxed{\mathfrak{D}_u(x, r; u(x)) \longrightarrow 0 \quad \text{as } r \downarrow 0.} \quad (25)$$

*Proof.* Put  $f = |M_2(Du)|^2 \in L^1$ . At almost every point,  $x$  is both a Lebesgue point of  $f$  and an  $L^2$  differentiability point of the Sobolev map  $u$ . Thus

$$\int_{B_r(x)} |u(y) - u(x)|^2 dy = O(r^5), \quad \int_{B_r(x)} |f(y) - f(x)| dy = o(r^3). \quad (26)$$

Since two points of  $\mathbb{S}^3$  have squared distance at most four,

$$\begin{aligned} \mathfrak{D}_u(x, r; u(x)) &\leq \frac{f(x)}{r^2} \int_{B_r(x)} |u - u(x)|^2 dy \frac{4}{r^2} \int_{B_r(x)} |f - f(x)| dy \\ &= O(r^3) + o(r) \longrightarrow 0. \end{aligned} \quad (27)$$

□

**Remark 5.2** (The remaining two-limit theorem). *Equation (25) is a genuine advance, but not the desired regularity gate. It first fixes  $u$  and then sends  $r$  to zero. The relaxation measure instead records a limit of  $u_j$ , so the required estimate has the stronger order*

$$\lim_{r \downarrow 0} \limsup_{j \rightarrow \infty} \frac{1}{r^2} \int_{B_r(x)} |u_j - q_{j,r}|^2 |M_2(Du_j)|^2 dy = 0 \quad \text{for } \mu\text{-a.e. } x, \quad (28)$$

for controlled centres  $q_{j,r} \in \mathbb{S}^3$ , or a quantitatively equivalent equiintegrability statement. A singular  $\mu$  may sit on the Lebesgue-null exceptional set; even an absolutely continuous base/mixed component is not eliminated until the recovery limit is tied to the fixed-map estimate. One natural route is a scale-invariant excess inequality of the form

$$\mathcal{X}(x, \vartheta r) \leq c \vartheta^\alpha \mathcal{X}(x, r) + c \mathfrak{D}_u(x, r; u_{x,r}), \quad 0 < \vartheta < \frac{1}{2}, \quad (29)$$

together with a sequence-uniform version of the second term. Ordinary Lebesgue differentiation neither exchanges these limits nor controls a singular support.

## 6 Revised closure chain and embargoes

The streamlined mainline is now:

1. **Frozen action—done.** AP-E11 (1)–(2) is final; no more lattice scans are authorized.
2. **Defect and fibre purification—partly done.** The Radon measure  $\mu$  is canonical, and its compact-fibre-phase-reducible component is zero. The base/mixed component remains.

3. **Hopf-base regularity—in progress.** Weighted cutoff decay is proved at Lebesgue-a.e. centres for each fixed map. It must be upgraded to the sequence-uniform  $\mu$ -a.e. estimate (28); only then may one conclude full  $\mu = 0$ , local graph recovery, regularity, and continuum isolation.
4. **Second variation—closed by policy.** A bosonic Riemann Hessian may be assembled only after classicality and isolation are theorems for the same action and representative.

The tangent Dirac/Callias operator on the same *prescribed* smooth Hopf base may be studied in parallel as an independent mathematical family. Such work does not authorize a determinant, a physical fermion assignment, or a degree-one portal. Determinant and portal promotion remain false until the classical bosonic background gate closes.

**Reproducibility.** Run `python3 route_f/code/verify_ap_e15_relaxation_defect_fibre_purification.py` from the repository root. The verifier checks the phase Hessian coefficients, the production convexity radius, the Hilbert defect identity, the  $r^3$  smooth weighted-cutoff scaling, and all fail-closed gates. It does not relax or scan a lattice.